

Course Syllabus

Chapter 1: Basics

- Computer Organization
- CPU
- Control Unit
- Input and Output Devices
- Monitor
- Software
- Application Software
- Operating System
- Translation System
- Software Development Activities
- Software Development Cycle
- IDEs
- Engineering Software
- Complexity Trade-off
- Software Engineering Goals
- Software Engineering Principles
- Abstraction
- Encapsulation
- Modularity
- Hierarchy
- Northern Timber Wolf Taxonomy
- OO Design and Programming
- Objects
- Binary Arithmetic
- Two's Complement
- Two's Complement Arithmetic

Chapter 2: The Fundamentals of C++

- Program Organization
- Object

- A First Program - Greeting.cpp
- Greeting Output
- Area.cpp
- Visual C++ IDE with Area.cpp
- Area.cpp Output
- Comments
- Fundamental C++ Objects
- Integer Object Types
- Integer Constants
- Decimal Constants
- Character Object Types
- Character Operations
- Character Constants
- Literal String Constants
- Floating-Point Object Types
- Floating-Point Constants
- Names
- Keywords
- Identifiers
- Definitions
- Examples
- Arithmetic Operators
- Integer Division
- Mod
- Operators and Precedence
- Operator Precedence
- Defining and Initializing

Chapter 3: Modifying objects

- Memory Depiction
- Assignment Statement
- Definition
- Consider
- Incrementing

- Const Definitions
- Assignment Conversions
- Nonfundamental Types
- Examples
- Class string
- EzWindows Library Objects
- Initialization
- An EzWindows Program
- An EzWindows Project File
- Sample Display Behavior
- RectangleShape Objects
- Another EzWindows Program
- Compound Assignment
- Increment and Decrement

Chapter 4: If Control Construct

- Boolean Algebra
- A Boolean Type
- Relational Operators
- Operator Precedence Revisited
- Conditional Constructs
- The Basic If Statement
- Example
- Sorting Two Numbers
- Semantics
- What is the Output?
- The If-Else Statement
- Finding the Max
- Selection
- An If-Else-If Statement
- A Switch Statement

Chapter 5: Iterative Constructs

- Averaging

- Determining Average Magnitude
- C++ Iterative Constructs
- While Syntax
- While Semantics
- Computing an Average
- Execution Trace
- Power of Two Table
- Better Way of Averaging
- Even Better Way of Averaging
- The For Statement
- Table Revisiting
- Displaying a Diagonal
- Sample Display
- Displaying Three Diagonals
- For Into While
- Counting Characters
- Counting All Characters
- File Processing
- Iteration Do's
- The Do-While Statement
- Waiting for a Proper Reply

Chapter 6: Libraries

- Functions
- Terminology
- Invocation Process
- Execution Process
- Function Prototypes
- Libraries
- Basic Translation Process
- Some Standard Libraries
- Library Header Files

Chapter 7: Programmer-defined Functions

- Function Definition
- Function Invocation
- Simple Programs
- Value Parameter Rules
- PromptAndRead()
- Sum()
- Problem
- Range.cpp
- Blocks and Local Scope
- Local Object Manipulation
- Name Reuse
- However, Don't Do This At Home
- Global Scope
- Don't Do This At Home Either
- Consider
- Using
- Doesn't do what we want!
- That would work!
- Reference Parameters
- Reconsider
- Extraction
- Getnum.cpp
- Constant Parameters
- Passing Constant Rectangles
- Default Parameters
- Function Overloading
- Random Numbers
- Uniform Random Numbers
- Stdlib Library
- Different Sequences
- Randseed.cpp
- Recursion
- Recursion Visualization

Chapter 8: The Class Construct

- Class Types
- Terminology
- Member Functions
- A Simple RectangleShape Class
- Simple RectangleShape Header File
- Simple RectangleShape
- Access Tests
- The RectangleShape Class

Chapter 9: Abstract Data Types

- Our Goal
- Golden Rule
- Another Golden Rule
- Abstract Data Type
- Rational Number Review
- Rational Attributes
- Rational Public Behaviors
- Non-Public Behaviors
- Auxiliary Behaviors
- Library Components
- MyProgram.cpp
- Producing MyProgram.exe
- Rational Header File Overview
- Class Rational Overview
- Rational Public Section
- Rational Protected Section
- Rational Private Section
- Auxiliary Operator Prototypes
- Auxiliary Operator Importance
- Const Power
- Rational Implementation
- Remember

- Specific Constructor
- Inspectors
- Numerator Mutator
- Denominator Mutator
- Addition Facilitator
- Multiplication Facilitator
- Insertion Facilitator
- Basic Extraction Facilitator
- Auxiliary Arithmetic Operators
- Auxiliary Insertion Operator
- Auxiliary Extraction Operator
- What's Happening Here?
- Copy Construction
- A Rational Copy Constructor
- Gang Of Three
- A Rational Assignment Operator
- Rational Destructor

Chapter 10: Arrays and Vectors

- Lists
- Array Terminology
- C++ Restrictions
- Basic Array Definition
- Example Definitions
- Subscripting
- Array Elements
- Array Element Manipulation
- Extracting Values For A List
- Displaying A List
- Smallest Value
- Preliminary Design
- Necessary Information
- A More Detailed Design
- Passing An Array

- Using ListMinimum()
- Remember
- Some Useful Functions
- Useful Functions Being Used
- Searching
- Sequential List Searching
- Example Invocation
- Sorting
- Common Sorting Techniques
- SelectionSort
- Complexity
- Vectors
- Standard Template Library
- STL Container Classes
- Vector Class Properties
- Example
- Some Vector Constructors
- Construction
- Vector Interface
- Function Examples
- Examples
- Overloading >>
- Iterators
- Iterator Operators
- SelectionSort Revisited
- QuickSort
- Picking The Pivot Element
- Decomposing Into Sublists
- Sorting Q W E R T Y U I O P
- InsertionSort
- Searching Revisited
- String Class Revisited
- Using GetWords()
- String Class As Container Class

- Reconsider A
- Counting o?s
- Explicit Two-Dimensional List
- Multi-Dimensional Arrays
- Memory Layout
- Identity Matrix Initialization
- Matrix Addition Solution

Chapter 11: Inheritance

- Think of a Bicycle
- Think of a Tandem Bike
- Think of a Racing Bike
- Think of a Mountain Bike
- Thinking About Bicycles
- Wouldn?t It Be Nice
- Inheritance
- Before Inheritance
- Shapes Hierarchy
- Class WindowObject
- WindowObject Constructor
- WindowObject Inspectors
- WindowObject Mutator
- Defining a Derived Class
- Declaring a Derived Class
- Implementing A Derived Class Constructor
- Implementing a Derived Class
- Basic Shapes
- TriangleShape
- EllipseShape
- RectangleShape
- TriangleShape::Draw()
- Using Shapes
- Fun with Shapes
- Cleaning Up

- Inheritance and Member Access
- Controlling Inheritance

Chapter 12: Pointers and Dynamic Objects

- Usefulness
- Categorizing Expressions
- Basics
- Syntax
- Memory Depiction
- Indirection Operator
- Address Operator
- Null Address
- Member Indirection
- Traditional Pointer Usage
- Constants and Pointers
- Differences
- General New Operation Behavior
- The Basic New Form
- Examples
- Another Basic New Form
- The Primary New Form
- Right Array For The Job
- Delete Operators
- Cleaning Up
- Dangling Pointer Pitfall
- Memory Leak Pitfall
- A Simple Dynamic List Type
- Sample IntList Usage
- IntList Definition
- IntList Definition (continued)
- Default Constructor
- Gang of Three Rule
- Why A Tailored Copy Constructor
- Tailored Copy Constructor

- Gang Of Three
- First Assignment Attempt
- Initial Implementation (Wrong)
- This Pointer
- Member Assignment Operator
- Accessing List Elements
- Stream Operators
- Beware of Friends
- Implementing Friend <<
- Proper << Implementation

Chapter 13: Templates and Polymorphism

- Polymorphic Functions
- Function Templates
- An Example Function Template
- Min Template
- Function Templates Facts
- Generic Sorting
- STL's Template Functions
- Class Templates
- A Generic Array Representation
- Homegrown Generic Arrays
- Auxiliary Operators
- Default Constructor
- Copy Constructor
- Destructor
- Member Assignment
- Inspector for Constant Arrays
- Nonconstant Inspector/Mutator
- Generic Array Insertion Operator
- Specific Array Insertion Operator
- Scenario
- Virtual Functions
- A Shape Class with a Virtual Draw

- Pure Virtual Function
- A Shape Abstract Base Class

Chapter 14: EzWindows API

- Event-based Programming
- EzWindows Coordinate System
- Class Position
- Position
- EzWindows Auxiliary Functions
- Class SimpleWindow
- Hello EzWindows
- Bitmaps
- Class BitMap
- Fun with Pictures
- Mouse Events
- Timer Events
- Timer Functions
- Example